



## History of death ray

Since time immemorial, we've tried to harness the sun into an effective weapon, but with little success <http://dgit.in/1orKWGW>



## Time travelling photons?

Can photons travel through time via a wormhole? What if they could? Hmm <http://dgit.in/1IK1wOl>

Space Age

## Building A Space Fence

The most obvious way to tackle the problem was devised as early as 1957 through implementing tracking systems which was in the form of NASA's Space Object Catalogue. But as the presence of space junk became a larger threat it is now maintained by NASA and the United States Department of Defense. Using their Space Surveillance Network technology systems they are able to maintain tracking information on objects as small as 5 centimeters within the Low Earth Orbit region and objects larger than 1 meters within the high altitude geosynchronous orbit region. In total the number of objects within their tracking parameters is only about 21,000 with only statistical guesses for the rest. And this scope is woefully inadequate

To make up for this lag U.S. government has been looking to create a more reliable solution which has popularly been known as the Space Fence. Unlike a physical wall, the Space Fence program would massively overhaul the existing tracking and surveillance systems to broaden their scope. The space fence is simply a high-frequency radar system that is capable of discovering even the smaller debris that are floating around in space.

The yet to be built system by the Americans will be awarded to either Lockheed Martin or Raytheon in 2014 and is expected to increase the tracking coverage by a factor of ten. Similar developments by the European Space Agency are also under way in the form of the Space Surveillance and Tracking system. The ESA's system will include data gathering from across the world in the form of radar and optical systems from national space agencies, defence departments and private institutions. The data gathered would be analysed for real time and planned guidelines for future space missions.

## Operation Clean Up

Even as these projects work their way towards implementation the real concern isn't on whether we will be able to track and adjust our space operations but how will we escape the inevitable reality of this space pollution. This problem is highly complex and layered; from setting up guidelines to-



The Sling Sat ballets across space from one debris to another using momentum as propulsion for minimum fuel use.

wards mitigating the addition of new debris all the way to figuring out a way of removing the space debris that are already in orbit. The multiple issues of technological capability, financial feasibility and ownership rights is a constant thorn in the side for anyone interested in taking on the challenge. The biggest one of which is figuring out a method of removing space junk as it already exists.

In 2012 a project designed by the Swiss Space Centre known as the Clean Space One was a step in the right direction. The Clean Space One vessel coupled with a defunct Swiss satellite and forced it to safely deorbit but the long term cost were calculated to be too high. Another one of the plans that is under consideration is the use of the commercial Space Infrastructure Servicing vehicle slated for launch in 2015. The SIS is meant to function as a fueling depot and service centre for communications satellites but is being designed with the ability to push defunct equipment towards "graveyard orbits" which would lead them to safely burnup in the higher atmosphere.

But as the costs of even these clean up mission comes at par with any other space mission they are not considered practi-

cal. To resolve this financial issue, a new approach has been devised known as the Space Sweeper with Sling-Sat or 4S. The 4S satellite is designed to grapple large space debris and eject them to a burn out orbit. The sequential movement from debris to debris provides the 4S with the momentum needed to move from target to target. But even with such innovation the issues of debris ownership still slow down progress. Simply put, a Swiss cleanup vehicle isn't allowed to even touch an American satellite without permission even if it's nothing more than trash. And permission is hard to come by from any country.

## Save Us From Ourselves

So even as technological innovation moves forward and undoes the consequences of past innovations we find ourselves back to the human element. Space is a vast arena where conflicts in nationality and politics aren't conducive to progress. The very ambition that sends us to space is now at risk and unless the countries of the world are able to set aside their political concerns and work together each passing day brings us closer to seeing the sky filled with junk rather than stars. ■